

GATE CS

Computer Science & Engineering

TREND-BASED MOCK PAPER

— 01 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 24 questions (37%)

Moderate: 33 questions (51%)

Difficult: 8 questions (12%)

GENERAL INSTRUCTIONS

GATE CS — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

3M [-0.66]

[General Aptitude]

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and 2?

- (A) $1 + 2$ is not a prime number.
- (B) 12 is not a prime number.
- (C) $1 + 2 + 1$ is a prime number.
- (D) $12 + 1$ is a prime number.

Q2.

2M [-0.66]

[General Aptitude]

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube? Note: The figures shown are representative.

- (A) A
- (B) B
- (C) C
- (D) D

Q3.

2M [-0.66]

[General Aptitude]

Consider the relationships among P, Q, R, S, and T: $\ddot{y}$ P is the brother of Q. $\ddot{y}$ S is the daughter of Q. $\ddot{y}$ T is the sister of S. $\ddot{y}$ R is the mother of Q. The following statements are made based on the relationships given above. (1) R is the grandmother of S. (2) P is the uncle of S and T. (3) R has only one son. (4) Q has only one daughter. Which one of the following options is correct?

- (A) Both (1) and (2) are true.
- (B) Both (1) and (3) are true.
- (C) Only (3) is true.
- (D) Only (4) is true.

Q4.

2M [-0.66]

[General Aptitude]

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.

- (A) ambivalence
- (B) satisfaction
- (C) resolve
- (D) revolve

Q5.

3M [-0.66]

[General Aptitude]

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is representative. Library 1st Main Road Canteen Physics Lab Hospital 5th Cross Road Chemistry Hostels Classrooms Lab N W E S

- (A) The library is located to the northwest of the canteen.
- (B) The hospital is located to the east of the chemistry lab.
- (C) The chemistry lab is to the southeast of physics lab.
- (D) The classrooms and canteen are next to each other.

Q6.

2M [no -ve]

[General Aptitude]

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

- (A) 25
- (B) 28
- (C) 30
- (D) 32

Q7.

2M [-0.66]

[General Aptitude]

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.

- (A) a; a; an
- (B) the; an; a
- (C) a; an; a
- (D) an; an; a

Q8.

2M [-0.66]

[General Aptitude]

Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work. How many attempts does each wizard make before coming to this conclusion, independently?

- (A) 24
- (B) 48
- (C) 16
- (D) 12

Q9.

2M [no -ve]

[General Aptitude]

Synonym of 'ubiquitous':

- (A) Rare
- (B) Omnipresent
- (C) Hidden
- (D) Unique

Q10.

2M [-0.66]

[Engineering Mathematics]

Derivative of x^3 at $x=2$:

- (A) 6
- (B) 12
- (C) 8
- (D) 4

Q11.

2M [no -ve]

[Engineering Mathematics]

Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

*Enter your answer here***Q12.**

2M [no -ve]

[Engineering Mathematics]

Mean of binomial $n=10, p=0.5$:

*Enter your answer here***Q13.**

2M [no -ve]

[Engineering Mathematics]

Probability of at least one head in 3 coin tosses:

- (A) $1/8$
- (B) $3/8$
- (C) $7/8$
- (D) $1/2$

Q14.

1M [-0.33]

[Engineering Mathematics]

Determinant of 2×2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q15.

1M [-0.33]

[Programming and Data Structures]

Infix to postfix uses:

- (A) Queue
- (B) Stack
- (C) Tree
- (D) Graph

Q16.

1M [no -ve]

[Programming and Data Structures]

What is the time complexity of inserting at the beginning of a linked list?

*Enter your answer here***Q17.**

1M [-0.33]

[Programming and Data Structures]

Avg hash lookup:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n^2)$

Q18.

1M [-0.33]

[Programming and Data Structures]

Inorder traversal of BST gives:

- (A) Sorted order
- (B) Reverse sorted
- (C) Level order
- (D) Random

Q19.

1M [no -ve]

[Algorithms]

LCS uses:

*Enter your answer here***Q20.**

1M [no -ve]

[Algorithms]

Merge sort space:

*Enter your answer here***Q21.**

1M [-0.33]

[Algorithms]

Kruskal uses:

- (A) DFS
- (B) BFS
- (C) Union-Find
- (D) Dijkstra

Q22.

1M [-0.33]

[Algorithms]

Time complexity of binary search:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(1)$

Q23.

1M [-0.33]

[Algorithms]

BFS uses:

- (A) Stack
- (B) Queue
- (C) Heap
- (D) Tree

Q24.

2M [no -ve]

[Algorithms]

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A) $O(V^2)$
- (B) $O(E \log V)$
- (C) $O(V \log V + E)$
- (D) $O(V \log V)$

Q25.

1M [no -ve]

[Algorithms]

Worst-case time complexity of merge sort:

*Enter your answer here***Q26.**

1M [no -ve]

[Operating Systems]

Round-robin uses:

*Enter your answer here***Q27.**

1M [-0.33]

[Operating Systems]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q28.

2M [-0.66]

[Operating Systems]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q29.

1M [-0.33]

[Operating Systems]

Inode contains:

- (A) File name
- (B) Metadata
- (C) File data
- (D) Directories

Q30.

1M [-0.33]

[Operating Systems]

TLB stands for:

- (A) Translation Lookaside Buffer
- (B) Time Local Buffer
- (C) Total Logic Block
- (D) None

Q31.

1M [-0.33]

[Operating Systems]

Which of the following is NOT a necessary condition for deadlock?

- (A) Mutual Exclusion
- (B) Hold and Wait
- (C) Preemption
- (D) Circular Wait

Q32.

1M [no -ve]

[Operating Systems]

Semaphore with value 1 is:

*Enter your answer here***Q33.**

1M [-0.33]

[Computer Networks]

Asymmetric encryption:

- (A) AES
- (B) DES
- (C) RSA
- (D) 3DES

Q34.

1M [-0.33]

[Computer Networks]

CSMA/CD is used in:

- (A) Token Ring
- (B) Ethernet
- (C) WiFi
- (D) Bluetooth

Q35.

1M [-0.33]

[Computer Networks]

Protocol to find MAC from IP:

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) ICMP

Q36.

1M [-0.33]

[Computer Networks]

Which protocol operates at the application layer of the TCP/IP model?

- (A) TCP
- (B) UDP
- (C) HTTP
- (D) IP

Q37.

1M [-0.33]

[Computer Networks]

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q38.

1M [-0.33]

[Computer Networks]

OSPF is:

- (A) Distance vector
- (B) Link state
- (C) Path vector
- (D) Hybrid

Q39.

1M [-0.33]

*[Digital Logic]*The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q40.

1M [no -ve]

[Digital Logic]

Full adder has how many inputs:

*Enter your answer here***Q41.**

1M [-0.33]

[Digital Logic]

Flip-flop stores:

- (A) 1 bit
- (B) 2 bits
- (C) 4 bits
- (D) 8 bits

Q42.

1M [-0.33]

[Theory of Computation]

Halting problem:

- (A) Decidable
- (B) Undecidable
- (C) Semi-decidable
- (D) None

Q43.

1M [no -ve]

[Theory of Computation]

PDA recognizes:

*Enter your answer here***Q44.**

1M [no -ve]

[Theory of Computation]

Which language is regular?

*Enter your answer here***Q45.**

1M [-0.33]

[Compiler Design]

3-address code operands:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q46.

1M [-0.33]

[Compiler Design]

Lexical analyzer outputs:

- (A) Tokens
- (B) Parse tree
- (C) Intermediate code
- (D) Assembly

Q47.

2M [no -ve]

[CS]

8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (in bits per second) of the DMA controller if 1% of the processor cycles are used for DMA?

*Enter your answer here***Q48.**

2M [no -ve]

[CS]

Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v , labelled c/X , where c is an input symbol or ϵ , X is a stack symbol, and α is a string of stack symbols, represents the fact that in state u , the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string α on the stack, and go to state v . In the initial configuration, the stack has only the symbol A in it. The PDA accepts by empty stack. $a/A \ b/A \ a/A/AA \ s \ b/A \ p \ /A/ \ /A/ \ q \ /A/ \ //$ Which one of the following options correctly describes the language accepted by P ?

Enter your answer here

Q49.

2M [no -ve]

[CS]

Let be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where , , , are Boolean variables?

- (A) $(, , (, ,)) = ((, ,), ,)$
- (B) $(, , (, ,)) = (, ,)$
- (C) $(, , (, ,)) = ((, ,) \text{ AND } (, ,))$
- (D) $(, ,) = (, (, ,), (, ,))$

Q50.

2M [no -ve]

[CS]

The pseudocode of a function fun() is given below: fun(int A[0,...,n-1]){ for i=0 to n-2 for j=0 to n-i-2 if $(A[j]>A[j+1])$ then swap A[j] and A[j+1] } Let [0, ... ,29] be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function fun() is called with [0, ... ,29] as argument, is _____. (Answer in integer) Organising Institute: IIT Roorkee Computer Science and Information Technology (CS1)

- (A) A
- (B) B
- (C) C
- (D) D

Q51.

2M [no -ve]

[CS]

Consider a Boolean expression given by $(, ,) = (3,5,6,7)$. Which of the following statements is/are CORRECT?

- (A) $(, ,) = (0,1,2,4)$
- (B) $(, ,) = + +$
- (C) $(, ,)$ is independent of input
- (D) $(, ,)$ is independent of input

Q52.

2M [-0.66]

[CS]

Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____

- (A) A
- (B) B
- (C) C
- (D) D

Q53.

2M [-0.66]

[CS]

A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded? Note: The figures shown are representative. (I) (II) (III) (IV)

- (A) A
- (B) B
- (C) C
- (D) D

Q54.

2M [-0.66]

[CS]

Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R1+R2$, which one of the following values of R1 and R2 gives an arithmetic overflow?

- (A) $R1 = 1011$ and $R2 = 1110$
- (B) $R1 = 1100$ and $R2 = 1010$
- (C) $R1 = 0011$ and $R2 = 0100$
- (D) $R1 = 1001$ and $R2 = 1111$

Q55.

2M [-0.66]

[CS]

Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T? Code Word M S-address R-address P Q S R T X DDS

- (A) P is 10:1 multiplexer; Q is 5:1 multiplexer; T is 2:1 multiplexer
- (B) P is 10:210 decoder; Q is 5:25 decoder; T is 2:1 encoder
- (C) P is 10:210 decoder; Q is 5:25 decoder; T is 2:1 multiplexer
- (D) P is 1:10 de-multiplexer; Q is 1:5 de-multiplexer; T is 2:1 multiplexer

Q56.

2M [-0.66]

[General Aptitude]

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and 2?

- (A) $1 + 2$ is not a prime number.
- (B) 12 is not a prime number.
- (C) $1 + 2 + 1$ is a prime number.
- (D) $12 + 1$ is a prime number.

Q57.

2M [-0.66]

[CS]

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let $\mathcal{F}$ be the set of functions from $\mathbb{N}$ to itself. For any two functions $f, g \in \mathcal{F}$, we define $(f+g)(n) = f(n) + g(n)$ for every number n in $\mathbb{N}$. Which of the following is/are CORRECT about the mathematical structure $(\mathcal{F}, +)$?

- (A) $(\mathcal{F}, +)$ is an Abelian group.
- (B) $(\mathcal{F}, +)$ is an Abelian monoid.
- (C) $(\mathcal{F}, +)$ is a non-Abelian group.
- (D) $(\mathcal{F}, +)$ is a non-Abelian monoid.

Q58.

2M [no -ve]

[CS]

An instruction format has the following structure: Instruction Number: Opcode destination reg, source reg-1, source reg-2 Consider the following sequence of instructions to be executed in a pipelined processor: I1: DIV R3, R1, R2 I2: SUB R5, R3, R4 I3: ADD R3, R5, R6 I4: MUL R7, R3, R8 Which of the following statements is/are TRUE?

Enter your answer here

Q59.

1M [no -ve]

[Programming and Data Structures]

Inorder traversal of BST gives:

Enter your answer here

Q60.

2M [no -ve]

[CS]

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right): initially after - 1* 2* 3* 4* 5* 4 1 6* 4 5 1* 3* 2* - 1 1 1 1 1 1 6 6 6 6 2 - - 2 2 4 4 4 4 4 4 1 1 1 1 - - - 3 3 5 5 5 5 5 5 3 3 3 The *-marked references cause page replacements. Which one or more of the following could be the page replacement policy/policies in use?

Enter your answer here

Q61.

2M [no -ve]

[CS]

Let $S = \{1, 2, 3, 4\}$. For S , let $\cdot$ be the product of symbols in modulo 7. We take $\epsilon = 1$, where ϵ is the null string. For example, $(124) = (1 \cdot 2 \cdot 4) \bmod 7 = 1$. Define $f(S) = \{S \mid (S) = 2\}$. The number of states in a minimum state DFA for $f(S)$ is _____. (Answer in integer)

Enter your answer here

Q62.

2M [no -ve]

[CS]

The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____. (answer in integer)

Enter your answer here

Q63.

2M [-0.66]

[CS]

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

- (A) The weak entity set MUST have total participation in the identifying relationship
- (B) The owner entity set MUST have total participation in the identifying relationship
- (C) Both weak and owner entity sets MUST have total participation in the identifying relationship
- (D) Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship

Q64.

2M [-0.66]

[CS]

"I put the brown paper in my pocket along with the chinks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past." (From G.K. Chesterton's "A Piece of Chalk") Based only on the information provided in the above passage, which one of the following statements is true?

- (A) The author of the passage carries a mirror in his pocket to reflect upon things.
- (B) The author of the passage had decided to write a poem on epics.
- (C) The pocket-knife is described as the infant of the sword.
- (D) Epics are described as too inconvenient to write.

Q65.

2M [-0.66]

[CS]

The value of the definite integral $\int_{-3}^{-1} \int_{-2}^{-1} \int_1^{21} (4x^2y - z^3) dz dy dx$ is . (Rounded off to the nearest integer)

CS GATE 2023 Computer Science and Information Technology (CS)

- (A) A
- (B) B
- (C) C
- (D) D